Prince Sultan University

College of Computer and Info Sciences / Department of Computer Science

Term 241

1st Semester 2024 - 2025

COURSE SYLLABUS

Mission Statement of the Bachelor of Computer Science Program(s): Provide high quality, computer science education to prepare top graduates through an environment that promotes innovative thinking, ethical behavior, lifelong learning, research, and service to the community.

1. Course number and name: CS175 Digital Logic and Computer Organization

0. Credits and contact hours: 3 credits and 4 contact hours (Lectures: 3 Tutorials:1)

0. Instructor's name: Dr. Abrar Wafa
a. Scheduled Office Hours:
Sunday, Tuesday, Wednesday: 11:00-12:00 PM
Monday: 11:00-01:00 PM
b. Office Location: **R102** – Building 101
c. Email: awafa@psu.edu.sa

0. Text book, title, author, and year
. Primary Text: "Logic and Computer Design Fundamentals", Morris Mano and Charles Kime. 4th Edition, Pearson New International Edition) Pearson , 2014

Other References:

• Stallings, Computer Organization and Architecture designing for performance, Pearson, 2009.

• "Essentials of Computer Architecture" by Douglas E. Comer, Pearson Prentice Hall, 2005.

Electronic Materials:
- http://www.prenhall.com/mano/
- www.howstuffworks.com
- www.wikipedia.com
- www.webopedia.com

Other learning material: Logisim, a graphical tool used to design, implement and test Logic circuits.

a. Learning Management System: Moodle available at https://lms.psu.edu.sa

0. Specific course information

. Brief description of the content of the course (catalog description):

This course explores computer organization and Digital logic. It describes how computers are organized and programmed at different abstraction levels. It covers a wide range of topics in computer hardware organization. This knowledge is a key factor in preparing the student to understand how computers work in subsequent courses.

a. Prerequisites or co-requisites: NA

b. Indicate whether a required, elective, or selected elective (as per Table 5-1) course in the program: Required as a Core course in BSCS program.

0. Specific goals for the course

. Specific outcomes of instruction. The student will be able to:

CLO1: Describe the basic theoretical concepts of digital systems like the binary system, Boolean algebra, and digital logic.

CLO2: Describe the role of each functional unit in performing the basic instruction execution cycle.

CLO3: Describe the organization of the basic computer system and its major functional units.

CLO4: Describe the organization and architecture of computer memory and buses.

CLO5: Simplify and optimize Boolean algebraic expressions representing basic combinational circuits.

CLO6: Design real life problems in logic design terminology and implement them using both combinational circuits and functional blocks such as decoders and multiplexers.

CLO7: Use of Digital Lab for design and analysis of Digital Circuits.

0. Brief list of topics to be covered

WEEK	UNIT/ TOPIC	CLO(s) Alignment	ASSIGNMENTS
1-4	**Data Representation** - Number Systems - Octal and Hexadecimal numbers - Decimal Representation - Complements - Subtraction of signed numbers - Arithmetic Addition - Arithmetic Subtraction - Subtraction with 2's complement	CLO1	Homework 1
5-8	**Introduction to Digital Logic Circuits** - Digital Computer - Logic Gates - Boolean Algebra - Standard Forms - Map simplification	CLO5, CLO7	Homework 2
9-10	**Combinational Logic Design** - Analysis Procedure - Design Procedure - Decoders - Multiplexers - Introduction to Sequential Circuits	CLO6, CLO7	Homework 3
11-12	**Basic Computer Organization & Design** - Design of a basic computer - Instruction Cycle - Instruction set architecture - RISC & CISC - Pipelining	CLO2, CLO3	
13-14	**Memory Organization** - RAM & ROM - Static & Dynamic memory - Types of DRAM - Byte addressing & Word Addressing - Memory hierarchy - Cache memory	CLO3, CLO4	Homework4
15	**Computer Buses** - Bus Arbitration	CLO4	

	- Distributed Arbitration		
16	**Review**	CLO2, CLO3, CLO7	

8. Weight of Assessments

- Attendance **5%**
- Assignments **5%**
- Lab **5%**
- Quizzes **20%** Best 2 Quizzes
  - Quiz 1: Wednesday, September 24 **(Break Time 12 – 1PM)**
  - Quiz 2: Wednesday, October 16 **(Break Time 12 – 1 PM)**
  - Quiz 3: Wednesday, November 20 **(Break Time 12 – 1 PM)**
- Major Exams 25%
  - One Major: Wednesday, October 9 **(Break time 12 – 1 PM)**
- Final Exam **40%**

9. Additional Information

Plagiarism and Academic Dishonesty: "Plagiarism can be defined as unintentionally or deliberately using another person's writing or ideas as though they are one's own. Plagiarism includes, but is not limited to, copying another individual's work and taking credit for it, paraphrasing information from a source without proper documentation, and mixing one's own words with those of another author without attribution. In addition, buying a paper or project, or downloading a paper from the Internet, and submitting them as your own are also plagiarism. The penalty for academic dishonesty will bring course expulsion and failure, or even suspension" (Academic Integrity and Syllabus Acknowledgement Form).

Attendance Policies: The University attendance policy will be strictly followed. Students are expected to attend all class sessions and be in class on-time. Attendance is taken during the first few minutes of the class. Missing a class session is a student's responsibility. Missed classes will not be repeated. A total of **14** absences may lead to denial grade DN. It is the student's responsibility to periodically check course website for course content, projects assignments, updates and notifications.

Exam Policies: Major exams are unified among all sections (if applicable) and their dates are announced from the onset of the semester. Arrangements with the Registration Office and the department are made before the beginning of the semester so that all sections will take the same exam and during the same time. It is not possible to reschedule any major exam. No student is allowed to take any assessment with another section unless there is a strong argument and in limited cases, to be approved by the course coordinator at least two days BEFORE the quiz date of the original student section. If any student missed an assessment, the makeup will be at the end of the semester and all the materials are included (comprehensive assessment). Makeup exam will only be approved in limited cases as stipulated in the university bylaws. Generally, the final exam includes all material covered during the semester (comprehensive).

Assignment/Project Policies: Students are expected to actively participate in class discussion and activities. Students are expected to solve the assignment and submit the solution during the concerned tutorial session. Late assignments are not accepted. However, lowest score assignment is dropped. It is the student's responsibility to check/test/verify/debug the code before submission. It is the student's responsibility to verify that all files have been uploaded to the LMS. For each project, instructor will provide 1-2 sample test-cases to verify the execution of your program. After an assignment/project has been graded, re-submission with an intention to improve an assignments scores will not be allowed. After the assignment/project has been graded, the instructor will post test-cases used for grading on the website. The Instructor has the right to share project execution reports that may have been auto-generated on the course website.

	No deduction	1 mark	2-mark Deduction	3-mark Deduction	4-mark Deduction	5-mark Deduction	DN
Twice weekly	Less than 3 absences	Total of 3 absences	Total of 4 absences	Total of 5 absences	Total of 6 absences	Total of 7 absences	Total of 8 absences
Three Times weekly	Total of or less than 2 absences	Total of 4 absences	Total of 6 absences	Total of 8 absences	Total of 10 absences	Total of 11 absences	Total of 12 absences
Four Times weekly	Total of or less than 3 absences	Total of 4 absences	Total of 7 absences	Total of 10 absences	Total of 13 absences	Total of 15 absences	Total of 16 absences
Six Times weekly	Total of or less than 4 absences	Total of 5 absences	Total of 9 absences	Total of 13 absences	Total of 21 absences	Total of 22 absences	Total of 23 absences